Simultaneous Surface Reduction and Self-Passivation via Ar-CH_4 Plasma for Low-Temperature Cu Bonding

Hoogwan Lee
Department of Semiconductor Engineering, Seoul National University of Science and Technology, Seoul, Korea
leehoogwan@naver.com

Byeongchan Go
Department of Semiconductor Engineering, Seoul National University of Science and Technology, Seoul, Korea
qudcks3464@seoultech.ac.kr

Sarah Eunkyung Kim
Department of Semiconductor Engineering, Seoul National University of Science and Technology, Seoul, Korea
eunkyung@seoultech.ac.kr

Abstract— **This study presents a one-step reductive Ar-CH_4 plasma pre-treatment for low-temperature Cu-to-Cu bonding. Optical emission spectroscopy (OES) revealed intense hydrogen radical and ion peaks, indicating a highly reductive plasma environment capable of removing surface oxides and activating Cu surfaces. Through Design of Experiments (DOE) optimization, high source power, low pressure, and elevated CH_4 flow rate were identified as optimal conditions, achieving effective reduction of 1–2 nm native Cu oxide and simultaneous deposition of a ~6 nm hydrogenated amorphous carbon (a-C:H) layer that prevented re-oxidation in air. Additional experiments were conducted by applying a small RF bias to the optimized plasma condition, which resulted in partial etching of the a-C:H layer rather than further improvement at the interface. Cross-sectional TEM and XPS analyses confirmed complete oxide removal and uniform a-C:H coverage under optimized plasma conditions. The influence of carbon layer thickness on bonding quality revealed a trade-off: thinner films enhanced Cu diffusion across the interface, improving bonding strength, but residual oxides were partially detected. Direct Cu-to-Cu bonding was carried out at 260 °C under 15 MPa for 1 h, followed by annealing at 200 °C for 1 h. The bonded interfaces exhibited minimal voids and improved mechanical integrity compared to untreated samples. Daisy-chain electrical measurements confirmed stable resistance comparable to pristine Cu, indicating that the a-C:H layer does not degrade electrical performance. These results demonstrate that the optimized Ar–CH_4 plasma treatment provides an effective one-step reduction and passivation route for reliable low-temperature Cu bonding applicable to advanced 3D stacking structures.**

Keywords : Reductive plasma treatment, Cu-to-Cu bonding, Self surface passivation

I. INTRODUCTION

The rapid advancement of artificial intelligence (AI) and high-performance computing has driven an increasing demand for high-speed and energy-efficient interconnects between logic and memory devices. High Bandwidth Memory (HBM) stacking technology, which integrates logic and memory through three-dimensional interconnections, has emerged as a solution to improve data bandwidth. Conventional packaging methods relying on solder bumps are limited by parasitic resistance, thermal stress, and reliability degradation as interconnect dimensions scale down. In addition, as stack height increases, challenges related to heat dissipation, signal delay, and mechanical integrity become more severe. [1-3]

Hybrid Cu-to-Cu bonding offers a promising alternative to solder-based interconnects due to its superior electrical conductivity, thermal performance, and scalability. Nevertheless, several issues hinder its practical adoption: (1) the formation of native Cu oxides that impede Cu diffusion, (2) non-uniform dishing and surface roughness after CMP, and (3) the requirement for bonding at elevated temperatures, which are incompatible with DRAM stacking processes due to thermal budget limitations. To address these challenges, surface pre-treatments based on plasma chemistry have gained increasing attention [3].

In this work, a novel one-step Ar-CH_4 plasma treatment is proposed as a potential approach to achieve simultaneous surface reduction and passivation for Cu-to-Cu bonding. The plasma environment generates both reducing hydrogen radicals and carbon-containing species, which are expected to remove native Cu oxides and form a thin hydrogenated amorphous carbon (a-C:H) layer that prevents re-oxidation prior to bonding. Although this mechanism suggests a promising route for low-temperature and reliable Cu interconnection, the detailed relationship between plasma parameters, interfacial chemistry, and resulting bonding performance has not yet been experimentally clarified. Therefore, this study investigates how plasma conditions influence Cu oxide reduction, a-C:H layer formation, and interfacial diffusion behavior. Through comprehensive surface and interface characterization—including XPS, TEM, TOF-SIMS, OES, and electrical daisy-chain measurements—the correlation between plasma-induced surface modification and bonding reliability is systematically evaluated.

II. EXPERIMENTAL

The overall experimental procedure is schematically illustrated in Fig. 1. An 8-inch Si wafer coated with a 700 nm

SiO_2 layer, a 50 nm Ti adhesion layer, and a 1 µm Cu film was used as the starting substrate. The Cu surface was planarized by chemical mechanical polishing (CMP), resulting in a surface roughness below 1 nm. Ar-CH_4 plasma pre-treatment was performed using an inductively coupled plasma chemical vapor deposition (ICP-CVD) system (Oxford PP100 Cobra 300, S-Fab@SeoulTech). Process optimization was carried out using a Design of Experiments (DOE) approach based on Response Surface Methodology (RSM), where ICP power, chamber pressure, and CH_4 flow rate were selected as variables. Additional experiments were conducted by applying a 30 W RF bias for 10 s prior to plasma exposure to promote further oxide reduction. To investigate the effect of carbon layer thickness, the plasma treatment time was varied to 15, 30, 45, and 60 s, yielding a-C:H films in the range of approximately 2-8 nm. After plasma treatment, Cu-to-Cu bonding was performed at 260 °C under 15 MPa for 1 h, followed by annealing at 200 °C for 1 h. Surface and interfacial characteristics were analyzed primarily by X-ray photoelectron spectroscopy (XPS, Thermo Nexsa) and transmission electron microscopy (Cs-corrected TEM, JEOL NEO ARM) to evaluate Cu oxide reduction, a-C:H layer structure, and the morphology of the bonded interfaces.

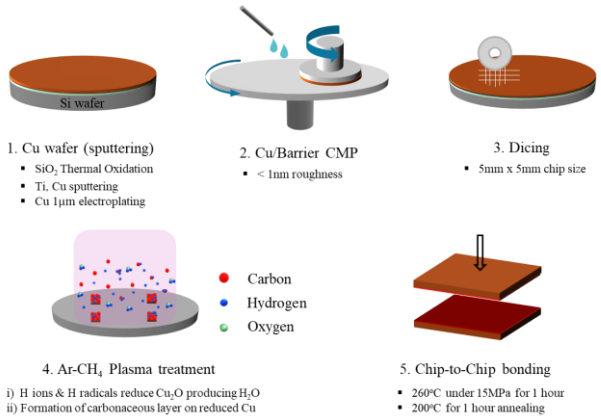


Figure 1. Overview of the Cu surface modification and bonding sequence using Ar-CH_4 plasma treatment.

III. Results and Discussion

A Design of Experiments (DOE) approach was conducted to evaluate the effects of ICP power, chamber pressure, and CH_4 flow rate on Cu oxide reduction and carbon layer formation. The optimized condition—high ICP power, low pressure, and high CH_4 flow—enhanced CH_4 dissociation and produced abundant hydrogen radicals and C_xH_y species, enabling simultaneous oxide reduction and uniform deposition of a hydrogenated amorphous carbon (a-C:H) film about 6–7 nm thick. The optimized plasma treatment produced an a-C:H layer approximately 6-7 nm thick on the Cu surface. XPS C1s analysis revealed that this layer was predominantly composed of sp^2-bonded carbon. In addition, O1s deconvolution showed the presence of OH-related components, suggesting the incorporation of hydrophilic functional groups. TOF-SIMS analysis further confirmed a high hydrogen concentration on the treated surface, demonstrating that the a-C:H layer formed by CH_4 plasma possesses good hydrophilicity. After bonding under this condition, cross-sectional TEM revealed a 15-20 nm carbon-rich interfacial region, indicating the formation of a carbon-to-carbon bonded layer generated by the rearrangement and coalescence of the opposing a-C:H surfaces during thermal compression.

The effect of carbon film thickness was also investigated using the optimized plasma treatment recipe. It was found that thinner a-C:H layers promoted interdiffusion of carbon into the underlying Cu during bonding. Finally, bonding quality was evaluated through shear testing, which confirmed that plasma-treated samples exhibited significantly higher bonding strength than untreated Cu, validating the effectiveness of the Ar-CH_4 plasma pre-treatment for robust low-temperature Cu-to-Cu bonding.

IV. Summary

A one-step reductive Ar–CH_4 plasma treatment was developed to simultaneously remove Cu oxides and deposit a protective a-C:H layer for low-temperature Cu-to-Cu bonding. OES, XPS, and TEM analyses revealed effective oxide reduction, uniform 6 nm a-C:H formation, and a thickness-dependent trade-off between bonding strength and residual oxides. Cu bonding at 260 °C achieved strong mechanical integrity and stable electrical performance, demonstrating the process's suitability for advanced packaging interconnect applications.

Acknowledgment

This work was supported by the Technology Innovation Program (Public-Private Joint Investment Semiconductor R&D Program [K-CHIPS] for nurturing high-level human resources; RS-2023-00236091, Development of hybrid bonding technologies for 3D packaging interconnects), funded by the Ministry of Trade, Industry and Energy (MOTIE, Korea; 1415187584). Additional support provided by a grant of the Basic Research Program funded by the Korea Institute of Machinery and Materials (grant number: NK254A, Development of Core Technologies for Advanced Chiplet Packaging Equipment). (Corresponding author: Sarah Eunkyung Kim).

References

[1] Lee Yoon-Gu; McInerney Michael; Joo Young Chang; Choi In Suk; Kim Sarah Eunkyung, "Copper Bonding Technology in Heterogeneous Integration" : Review paper, Electronic Materials Letters (2024).

[2] H. Park and S. E. Kim, "Two-Step Plasma Treatment on Copper Surface for Low-Temperature Cu Thermo-Compression Bonding," IEEE Trans Compon Packaging Manuf Technol, vol. 10, no. 2, pp. 332–338, Feb. 2020, doi: 10.1109/TCPMT.2019.2928323.

[3] A. Zhou et al., "Research progress of hybrid bonding technology for three-dimensional integration," Apr. 01, 2024, Elsevier Ltd. doi: 10.1016/j.microrel.2024.115372.